

Retrieval Systems

Retrieval systems find relevant documents from a large collection based on user queries. Modern retrieval systems go beyond simple keyword matching to use semantic understanding. The retrieval process typically involves encoding the query into embeddings, searching a vector database for similar vectors, and ranking results by relevance. Key metrics include precision, recall, and mean reciprocal rank. Advanced techniques include hybrid search combining keyword and semantic approaches, re-ranking models that refine initial results, and query expansion to improve recall. Effective retrieval is the foundation of successful RAG implementations.